

Gautam K Das

Tasks:

Task 1: SQL-Based Traffic Analysis

Using the Metro Interstate Traffic Volume dataset:

1.1 Load the dataset

- Load the comma-delimited traffic CSV file into **SQLite**.
- Verify that the data has been loaded correctly.

Loaded Metro_Interstate_Traffic_Volume.csv into a SQLite table traffic_volume (schema in 01_schema.sql) using DB Browser for SQLite's CSV import feature. Verified the load with the following checks, all matching the source CSV

Row count: 48,204 (via SELECT COUNT(*) FROM traffic_volume;)

Column count: 9, with correct data types (temp/rain_1h/snow_1h as REAL, clouds_all/traffic_volume as INTEGER)

Min/max traffic_volume: 0 and 7,280, matching the source data

Table: traffic_volume									
	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	None	288.28	0.0	0.0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
2	None	289.36	0.0	0.0	75	Clouds	broken clouds	2012-10-02 10:00:00	4516
3	None	289.58	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 11:00:00	4767
4	None	290.13	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 12:00:00	5026
5	None	291.14	0.0	0.0	75	Clouds	broken clouds	2012-10-02 13:00:00	4918
6	None	291.72	0.0	0.0	1	Clear	sky is clear	2012-10-02 14:00:00	5181
7	None	293.17	0.0	0.0	1	Clear	sky is clear	2012-10-02 15:00:00	5584
8	None	293.86	0.0	0.0	1	Clear	sky is clear	2012-10-02 16:00:00	6015
9	None	294.14	0.0	0.0	20	Clouds	few clouds	2012-10-02 17:00:00	5791
10	None	293.1	0.0	0.0	20	Clouds	few clouds	2012-10-02 18:00:00	4770
11	None	290.97	0.0	0.0	20	Clouds	few clouds	2012-10-02 19:00:00	3539

1.2 Analyse annual traffic trends

Compare the **total yearly traffic volume** for:

- 2012
- 2013
- 2014
- 2015
- 2016
- 2017

Total traffic volume by year (after removing duplicate-timestamp rows, explained below):

2012: 6,785,754

2013: 24,139,878

2014: 14,718,915

2015: 11,706,145
2016: 25,032,183
2017: 29,420,221

Note: I found that 5,445 timestamps in this dataset appear 2-3 times each, because the weather feed sometimes logs more than one observation within the same hour. I confirmed traffic_volume is identical across these duplicate rows, so I removed duplicates (keeping one row per unique hour, via the hourly_traffic view in 02_annual_trends.sql) before summing. Without this step, the totals above would be inflated.

Then:

- Identify years with increases or decreases.

2013: increase vs. 2012
2014: decrease vs. 2013
2015: decrease vs. 2014
2016: increase vs. 2015
2017: increase vs. 2016

- Calculate the change between years where appropriate.

	year	hours_recorded	total_traffic_volume	total_change_vs_prev_year	total_pct_change_vs_prev_year	avg_traffic_volume	avg_change_vs_prev_year	avg_pct_change_vs_prev_year
1	2012	2103	6785754	NULL	NULL	3226.7	NULL	NULL
2	2013	7294	24139878	17354124	255.74	3309.6	82.9	2.57
3	2014	4501	14718915	-9420963	-39.03	3270.1	-39.4	-1.19
4	2015	3593	11706145	-3012770	-20.47	3258.0	-12.1	-0.37
5	2016	7838	25032183	13326038	113.84	3193.7	-64.3	-1.98
6	2017	8713	29420221	4388038	17.53	3376.6	182.9	5.73

- Explain at least **two meaningful observations** about traffic volume trends.
 - o The totals above are heavily shaped by how much data exists for each year, not just real traffic demand. The percentage of the year actually covered by hourly records ranges from 23.9% (2012) to 99.5% (2017). This is why 2013 looks like a 255% jump over 2012, as the year 2012 only has data from October 2 onward (~2 mths). Thus, not a genuine traffic surge.
 - o A fairer comparison is average traffic volume per recorded hour, since its agnostic to how much data exists for each year. From that perspective, 2012-2015 stayed in a narrow band (about 3,227 to 3,310 vehicles/hour). However, 2016 was the lowest of all six years (3,193.7/hour) despite having good coverage (89.2%). 2017 then reached the highest average of any year (3,376.6/hour, +5.73% over 2016), and since 2017 also has the most complete coverage (99.5%), this looks like a genuine increase in traffic demand rather than a data artifact.

1.3 Analyse temperature around holidays

Compare temperature patterns for **2015, 2016 and 2017** during:

- New Year's Day

- Labor Day

Holiday	Year	Avg Temp (°C)	Avg Temp (K)	Hours Recorded	Avg Traffic Volume
New Year's Day	2015	No data*	No data*	0	No data*
New Year's Day	2016	-6.06	267.09	18	1832.00
New Year's Day	2017	-3.06	270.09	24	2127.60
Labor Day	2015	22.17	295.32	24	2430.30
Labor Day	2016	21.86	295.01	24	2151.50
Labor Day	2017	18.14	291.29	24	2395.20

* **Note:** New Year's Day 2015 could not be included in this comparison as there are zero hourly records for that date due to large gap between 2014-08-08 and 2015-06-11 (~10mths). As a result, I have marked 2015 New Years day as 'No data'.

	label	hours_recorded	min_temp_K	max_temp_K	avg_temp_K	avg_traffic_volume
1	New Years Day 2015 (Jan 1)	0	NULL	NULL	NULL	NULL
2	New Years Day 2016 (Jan 1)	18	263.99	271.98	267.09	1832.0
3	New Years Day 2017 (Jan 1)	24	267.71	271.9	270.09	2127.6
4	Labor Day 2015 (Sep 7)	24	291.85	299.6	295.32	2430.3
5	Labor Day 2016 (Sep 5)	24	291.88	297.99	295.01	2151.5
6	Labor Day 2017 (Sep 4)	24	284.93	296.22	291.29	2395.2

Identify notable year-on-year changes and discuss whether these changes appear relevant to traffic conditions.

Notable YoY Changes:

New Year's Day: 2016 - 2017 warmed by +3.00 degrees C (only one year-on-year comparison is possible, since 2015 has no data).

Labor Day: 2015 - 2016 was essentially flat (-0.31 degrees C). 2016 - 2017 noted a drop of -3.72 degrees C, the largest change observed in this analysis.

"Whether these changes appear relevant to traffic conditions":

- For New Year's Day, average traffic volume also rose alongside the temperature increase (2016: 1,832.0 to 2017: 2,127.6, +16.1%), which is directionally consistent with warmer weather coinciding with more traffic. As indicated in the previous task 2017 had the highest overall traffic of any year in the given dataset and the temperature change can be considered as one of the factors rather than the only factor contributing to a YoY growth.
- For Labor Day, the relationship breaks down, as the average temperature changed only marginally (a decrease of -0.31 degrees C) between 2015 and 2016. However, traffic fell by 11.5% (2,430.3 - 2,151.5). Thereafter between 2016 and 2017 temperature dropped sharply by ~ - 3.72 degrees C (i.e. relatively cooler) but traffic registered an increase of 11.3% (2,151.5 to 2,395.2) which is contrary to what one might expect if cooler weather suppressed

travel. So, this mismatch indicates that temperature is not a strong driver of changes in traffic for Labor Day.

Task 2: Descriptive Statistics and Correlation

Using the traffic dataset, calculate:

2.1 Traffic volume statistics

- Mean
- Median
- Standard deviation
- Variance
- Range

Mean	3,290.65
Median	3,427.00
Std Dev	1,984.77
Variance	3,939,323.50
Minimum	-
Maximum	7,280.00
Range	7,280.00

Interpret what these statistics tell you about traffic volume and its variability.

Mean vs Median: The mean (3,290.65) is noticeably lower than the median (3,427.00), a gap of about 136 vehicles. Since the mean is below the median, the distribution of traffic volume leans toward lower values. There is a cluster of low traffic hours, mainly overnight and early morning when volume drops close to zero, pulling the average down, while most hours sit at moderate to high volumes clustered nearer the median.

Std Deviation & Variance: The standard deviation (1,984.77) is large relative to the mean, and looking at the coefficient of variation at ~60.3% of the mean value ($1,984.77 / 3,290.65$ is ~ 60.3%) indicates substantial variability i.e. traffic volume swings widely rather than staying close to its average. The variance (3,939,323.50) conveys the same spread in squared units, so it is mainly useful as an input to other calculations rather than something to interpret directly on its own.

Range: The range (7,280, from a minimum of 0 to a maximum of 7,280 vehicles per hour) confirms this. The corridor goes from essentially empty at its quietest to fully saturated at its busiest within the same dataset.

Looking at the above statistics as a cohort the trend seems consistent. Traffic volume is not stable throughout the day. It depends heavily on the time of day, with a wide range between quiet overnight hours and peak commuting hours. A single average figure hides most of what is actually happening, and any predictive model needs to account for this time driven swing rather than treating traffic volume as roughly constant.

2.2 Correlation analysis

Calculate the correlation coefficient between:

- Temperature
- Traffic volume

Pearson correlation coefficient (temperature vs. traffic volume), calculated across all 40,575 unique hourly records in the dataset **$r = 0.14$ (unrounded: 0.1369)**. Because a Pearson correlation is unaffected by a linear rescaling of either variable, this value is identical whether temperature is expressed in Kelvin (as available in the file) or converted to Celsius.

Note: 10 rows record temperature as exactly 0 K (absolute zero), and a further 78 rows record exactly 276.793 K — both look like placeholder/default values rather than real sensor readings (0 K is physically impossible for Minneapolis–St Paul (unless it is created in a lab), and 276.793 K recurs suspiciously often, always on the "extra" row of an hour that already has a duplicate weather entry). Recomputing the coefficient with these 88 rows (0.22% of the data) excluded gives $r = 0.1391$, which still rounds to 0.14, so this data-quality issue does not change the result.

Interpret the direction of the relationship.

Direction: The coefficient is positive ($r = 0.14 > 0$), so temperature and traffic volume move in the same direction on average: hours with higher temperatures tend to have slightly higher traffic volume, and hours with lower temperatures tend to have slightly lower traffic volume. The relationship is not negative, **so there is no evidence here that heat suppresses driving.**

Interpret the strength of the relationship.

Strength: By conventional benchmarks for correlation strength (roughly: <0.20 very weak, 0.20 – 0.39 weak, 0.40 – 0.59 moderate, 0.60 – 0.79 strong, ≥ 0.80 very strong), $r = 0.14$ falls in the very weak range that indicates close to no linear relationship at all. Squaring the coefficient gives $R^2 \approx 0.019$, meaning temperature alone statistically accounts for under 2% of the hour-to-hour variation in traffic volume. This lines up with the sql task finding that traffic volume is dominated by time-of-day/commuting patterns rather than a single continuous driver such as temperature.

Explain why correlation does not necessarily imply causation.

Correlation vs. causation: A correlation coefficient only measures how closely two variables move together; it cannot by itself show that one causes the other. Two variables can be correlated because a third, unmeasured variable drives them both (a confounder), because the relationship actually runs in the opposite direction, or simply by chance. In this dataset, a plausible confounder is time (hour of day and season): traffic volume follows a strong commuting-driven daily cycle that has nothing to do with the outdoor temperature at that moment, while temperature follows its own daily and seasonal cycle. The two series can drift together somewhat simply because both are shaped by the calendar, without temperature having any direct influence on how many vehicles are on the road. For example, a warm summer rush hour and a cold winter rush hour can both be "high traffic" for scheduling reasons alone. Given how weak this correlation already is ($R^2 < 2\%$), it would be a mistake to conclude from it that warmer weather causes more people to drive; the data only supports saying the two are weakly, positively associated.

Overall, temperature has only a marginal linear association with traffic volume in this dataset. It is directionally positive but too weak to treat as a meaningful driver of traffic on its own, reinforcing that time-based factors (hour, day of week, holiday status) are far more important explanatory variables than weather.

Task 3: Probability and Congestion Analysis

Congestion is defined for this task as an hour where **traffic_volume > 5,500 vehicles**.

3.1 Basic probability

Calculate:

- $P(\text{Congestion})$
- $P(\text{Clear Weather})$
- $P(\text{Congestion AND Clear Weather})$

Method: All three probabilities are empirical relative frequencies computed over the same 40,575-row deduplicated, one-row-per-hour dataset used throughout Tasks 1.2, 2.1 and 2.2 (so an hour with multiple weather log entries is not double-counted). "Clear Weather" uses the weather_main column's exact category "Clear" (one of 10 categories in the data: Clouds, Clear, Rain, Drizzle, Mist, Fog, Thunderstorm, Haze, Snow, Squall, Smoke) rather than the free-text weather_description field, since weather_main gives a single unambiguous label per hour.

P(Congestion): 6,107 of 40,575 hours have traffic_volume > 5,500, so $P(\text{Congestion}) = 6,107 / 40,575 \approx 0.1505$ (15.05%).

P(Clear Weather): 13,366 of 40,575 hours are logged with weather_main = "Clear", so $P(\text{Clear Weather}) = 13,366 / 40,575 \approx 0.3294$ (32.94%).

P(Congestion AND Clear Weather): 1,762 of 40,575 hours are both congested and clear, so $P(\text{Congestion AND Clear Weather}) = 1,762 / 40,575 \approx 0.0434$ (4.34%).

3.2 Conditional probability

Calculate:

- $P(\text{Clear Weather} \mid \text{Congestion})$
- $P(\text{High Temperature} \mid \text{Congestion})$, where high temperature > 292K
- Determine whether weather conditions and congestion can be considered independent using $P(A \cap B) = P(A) \times P(B)$
- Calculate the odds ratio of congestion in clear versus cloudy weather.

Conclude with a short explanation of what the probability analysis suggests about the relationship between weather and congestion.

P(Clear Weather | Congestion): 1,762 of the 6,107 congested hours were logged as Clear, so $P(\text{Clear Weather} \mid \text{Congestion}) = 1,762 / 6,107 \approx 0.2885$ (28.85%).

P(High Temperature | Congestion): 1,699 of the 6,107 congested hours have temperature above 292K, so $P(\text{High Temperature} \mid \text{Congestion}) = 1,699 / 6,107 \approx 0.2782$ (27.82%).

Independence check: if Congestion and Clear Weather were independent, $P(\text{Congestion}) \times P(\text{Clear Weather})$ would equal $P(\text{Congestion AND Clear Weather})$. Here $P(\text{Congestion}) \times P(\text{Clear Weather}) = 0.1505 \times 0.3294 \approx 0.0496$ (4.96%), while the actual joint probability is 0.0434 (4.34%). These two values are not equal, so Congestion and Clear Weather cannot be considered independent.

Odds ratio: among clear hours, 1,762 were congested and 11,604 were not, giving odds of $1,762 / 11,604 \approx 0.1518$. Among cloudy hours, 2,587 were congested and 12,536 were not, giving odds of $2,587 / 12,536 \approx 0.2064$. The odds ratio of congestion in clear weather versus cloudy weather is $0.1518 / 0.2064 \approx 0.7358$.

Conclusion: Overall the probability analysis suggests that weather has only a mild relationship with congestion. Congestion is somewhat less likely during clear conditions than the independence assumption would predict, and the odds of congestion in clear weather are about 26 percent lower than in cloudy weather, since the odds ratio is below 1. This points in the same direction as the weak positive correlation found between temperature and traffic volume in Task 2.2, but the effect size is small in both cases. Weather appears to shift the likelihood of congestion only slightly rather than acting as a major driver of it, so factors such as time of day and day of week are still likely to matter more than weather conditions alone.

Task 4: Power BI Traffic Intelligence Dashboard

Import the traffic CSV file into **Power BI Desktop** and use **Power Query Editor** to prepare the data.

4.1 Data quality and preparation

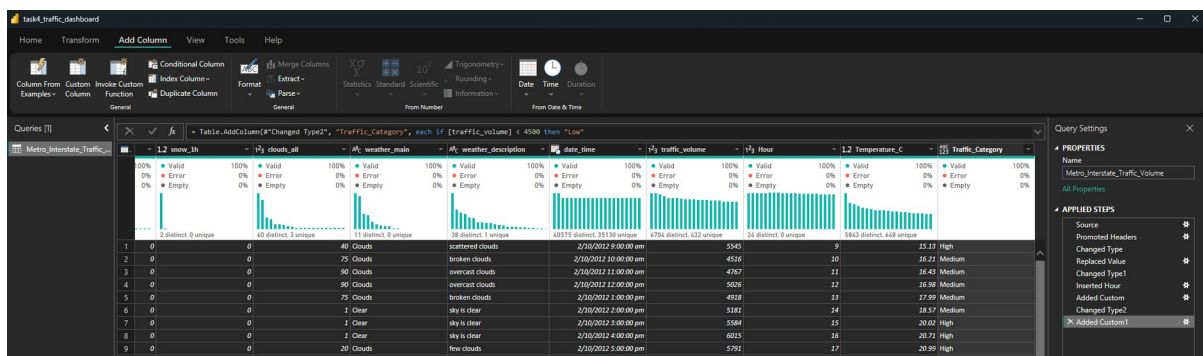
Perform the following:

- Determine the number of rows and columns.
- Identify null/missing values and the columns affected.
- Check whether the data types are appropriate.
- Extract **Hour** from DateTime.
- Create **Temperature in Celsius**.
- Create a **Traffic Category**:

Traffic Volume	Category
< 4,500	Low
4,500–5,500	Medium
> 5,500	High

Steps Followed as below:

1. Row and column count: the imported CSV has 48,204 rows and 9 columns (holiday, temp, rain_1h, snow_1h, clouds_all, weather_main, weather_description, date_time, traffic_volume).
2. Null and missing values: Power Query's Column quality view initially reported 0% Empty for every column, which looks like no missing data at all. However holiday stores the literal text None for non holiday hours rather than a true blank, in 48,143 of 48,204 rows (99.87%). This is not detected as missing until the text is explicitly replaced. Applying Replace Values (None to blank) on holiday raised its reported Empty percentage to 99%, correctly reflecting that almost every hour has no holiday.
3. Data type check: date_time, clouds_all and traffic_volume were auto detected with appropriate types (Date/Time and Whole Number). However temp, rain_1h and snow_1h were auto detected as Text rather than a numeric type, which would have prevented correct aggregation and charting. These three were changed to Decimal Number. As a side effect, rain_1h's reported distinct value count dropped from 372 to 30, since several text representations of the same number (such as 0 and 0.0) had been counted as different values before the fix, a further sign the original type was wrong.
4. Hour: extracted from date_time using Time.Hour([date_time]), producing a Whole Number column ranging 0 to 23.
5. Temperature in Celsius: created as a Custom Column named Temperature_C using the formula [temp] minus 273.15. Power Query initially left this column as the generic Any type, so it was explicitly changed to Decimal Number.
6. Traffic Category: created as a Custom Column named Traffic_Category using nested if logic so every row falls into exactly one non overlapping band: Low below 4,500, Medium from 4,500 up to and including 5,500, and High above 5,500, the same threshold used to define Congestion in Task 3. Spot checking against known traffic_volume values confirmed correct results, for example 5,545 correctly categorised as High and 4,516 as Medium.

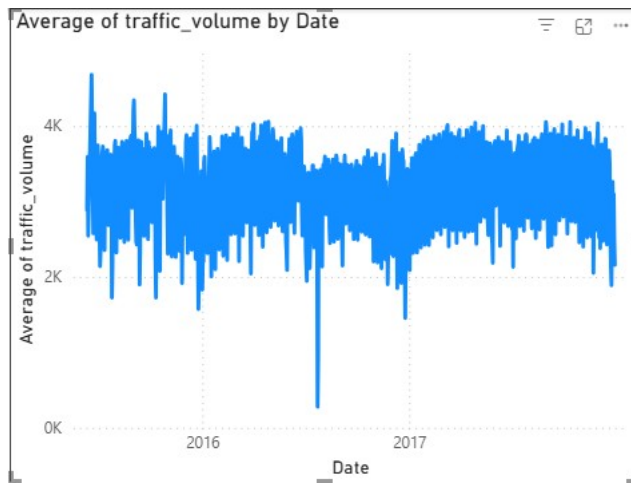


4.2 Dashboard analysis

Create visualisations showing:

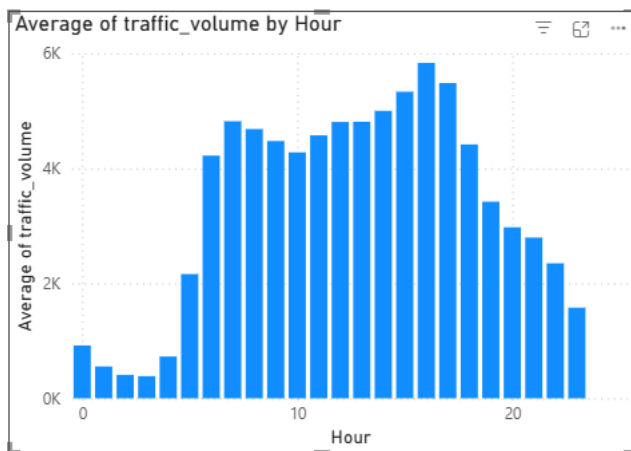
A. Daily traffic trends

- Show daily average traffic trends for 2015, 2016 and 2017.



B. Hourly traffic patterns

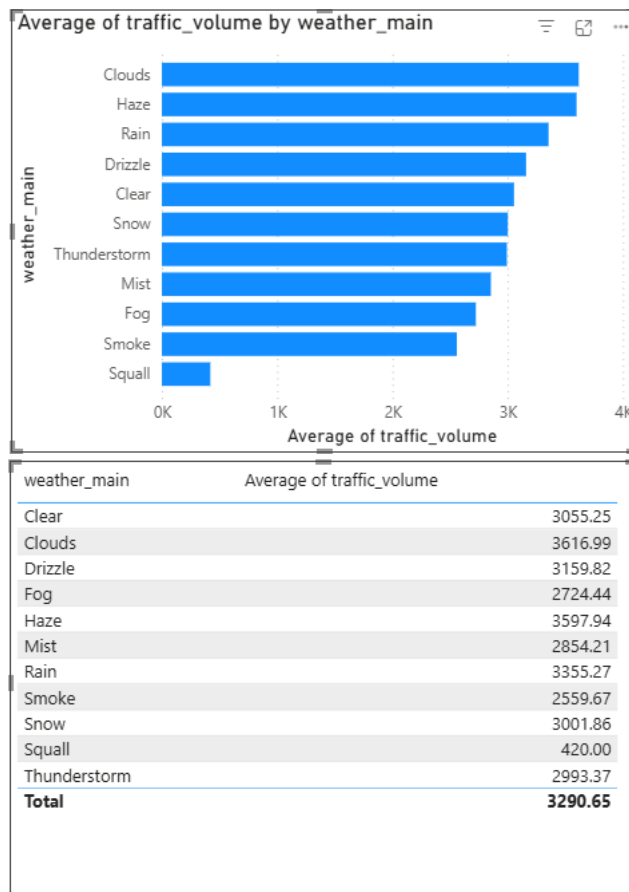
- Create a column chart showing average traffic by hour for 2017.



C. Weather impact

Analyse average traffic by weather condition and answer:

- Which weather condition has the highest average traffic?
- Which has the lowest?
- What is the difference between the highest and lowest average traffic?



Average traffic volume varies considerably across weather conditions, ranging from a high of 3,616.99 under **Clouds** to a low of 420.00 under **Squall**, a difference of 3,196.99 vehicles. This means average traffic during squalls is less than 12% of average traffic under cloudy conditions.

Clouds has the highest average traffic volume (3,616.99), closely followed by Haze (3,597.94) and Drizzle (3,159.82). This suggests that overcast or mildly adverse conditions don't meaningfully suppress travel, commuters largely continue their normal patterns under cloudy or hazy skies.

Squall has the lowest average traffic volume by a wide margin (420.00), far below every other condition, the next-lowest, Smoke, still averages 2,559.67. This sharp drop-off likely reflects two compounding factors: squalls are brief, severe weather events that actively discourage or prevent travel, and they occur rarely in the dataset, meaning this average is based on a much smaller number of hourly records than conditions like Clouds or Clear. As a result, the Squall figure should be interpreted cautiously rather than treated as a robust estimate of typical traffic behavior under that condition, the true underlying pattern is likely "severe weather sharply reduces travel," with the extremity of the number partly an artifact of limited data.

Overall, common, low-severity weather conditions (Clouds, Haze, Drizzle, Clear, Snow) cluster in a relatively high-traffic band (roughly 3,000-3,620), while rarer or more severe

conditions (Mist, Rain, Thunderstorm, Smoke, Fog, and especially Squall) trend toward lower averages, consistent with reduced travel during more disruptive weather.

D. Temperature and traffic

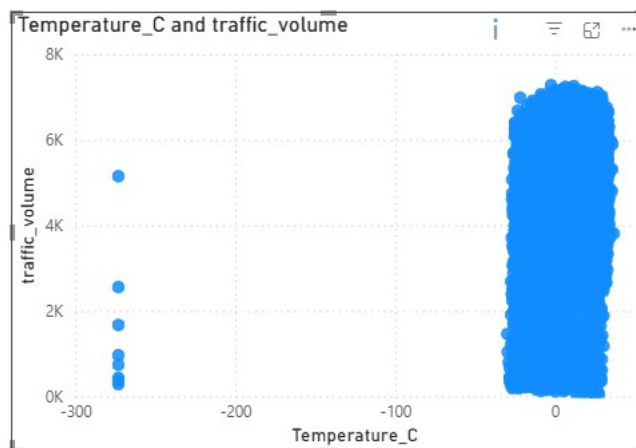
Create a scatter plot of:

- Temperature vs. traffic volume.

Discuss:

- Whether a visible relationship exists.
- The temperature range associated with higher traffic.

Any notable outliers.



The scatter plot of Temperature (°C) against traffic volume shows no strong linear relationship between the two variables. Rather than a clear upward or downward trend, the points form a dense vertical band across the real temperature range, indicating that both **high and low traffic volumes occur across nearly the full spectrum of observed temperatures. This suggests temperature is not a major driver of traffic volume on its own;** other factors, most likely time of day and day of week, are more influential in explaining the low-traffic observations scattered throughout the range.

Higher traffic volumes (roughly 5,000-7,500 vehicles) are most densely concentrated between approximately -30°C and +35°C, which represents essentially the full range of genuine weather conditions in the dataset. Within that band, there isn't a specific narrower sub-range that stands out as producing disproportionately higher traffic, high volumes appear consistently across cold, mild, and warm conditions alike, reinforcing that temperature alone doesn't meaningfully constrain when peak traffic occurs.

The most notable outliers are a small cluster of points sitting far to the left of the main distribution, at approximately -270°C. This is not a real temperature, it corresponds to temp

values of 0 Kelvin in the raw data, which is a sensor or data-logging error rather than an actual weather reading (**0 Kelvin is absolute zero, physically impossible at Earth's surface**). These erroneous records span a range of traffic volumes (roughly 400 to 5,200), confirming they're a data quality artifact unrelated to genuine temperature effects rather than a real cold-weather pattern. This cluster should be treated as a known data issue in the analysis and, ideally, excluded or corrected in any downstream modelling to avoid distorting temperature-based conclusions.

4.3 KPI cards and filters

Create KPI cards showing:

- Total hours analysed
- Average traffic volume
- Average temperature

Add slicers for:

- Hour range
- Weather condition
- Traffic category

